Volume 3: Screen-by-Screen Operator Manual and UI Guide

1. Introduction: Mastering Every Screen, Button, and Metric

During technical evaluations, hackathons, and municipal reviews, judges frequently point at specific UI elements, badges, charts, and buttons and ask:

- *"What does this specific button do when you click it?"*
- *"What does 'Gain vs Base' mean in this table?"*
- *"What is a 'Skill Score' and how is it calculated?"*
- *"Why does the model use XGBoost here instead of a neural network?"*
- *"Where did this number come from, and what physical process produced it?"*

This volume provides an exhaustive, granular reference manual for all 11 views across the platform. For every single screen, you will find:

1. The operational purpose of the page.
2. The complete technical stack powering that view.
3. Every button, toggle, and dropdown with its exact click action and backend handler.
4. Every number, gauge, and badge with its physical unit, normal range, and meaning.
5. The behind-the-scenes engineering algorithms (including Gain vs Base, Forecast Skill Score, Persistence Baselines, XGBoost multi-horizon inference, and Conformal Prediction).
6. Step-by-step operator demonstration instructions.

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2. Screen 0: Public Landing Page (`#landing`)

2.1 Operational Purpose

The public gateway and executive showcase. It provides city commissioners, mayors, transport planners, and the public with a clear, high-level overview of the digital twin mission, demonstrating the difference between uncoordinated city operations and digital twin-enabled coordination before entering the live console.

2.2 Technical Stack Used on This Screen

- **React 18 + TypeScript 5.5:** Renders the responsive glass-morphism layout.
- **Tailwind CSS:** Responsive typography, dark/light theme tokens, and grid layouts.
- **Lucide React:** Accessible vector iconography (zero emojis).
- **HTML5 Canvas & SVG:** Powers the interactive before-and-after corridor comparison diorama and the 4-tier pipeline animation.

2.3 UI Controls, Buttons, and Navigation

- **"Dashboard" / "Enter Operations Console":** Located in the top header and hero banner. Clicking this changes the application route hash to `#operations`, launching the primary Corridor Operations Console.
- **"Sign In":** Navigates to `#signin` (`AuthPageView`), allowing certified officers to log in with specific role privileges.
- **"Contact Us" / "Request Access":** Navigates to `#provisioning`, allowing external municipal agencies to submit corridor onboarding requests.
- **Interactive Before-and-After Slider:** An interactive split-screen comparison slider. Dragging the slider handle horizontally reveals simulated corridor traffic flow before digital twin coordination (congested, red segments, long queues) versus after coordination (smooth green progression, synchronized platoons).
- **Section Navigation Anchors:** "Telemetry", "Physical Assets", "Impact", "Pipeline", "Governance", "Operations", "Provenance", and "FAQ" smoothly scroll the page to the respective executive briefing cards.

2.4 Screen Numbers and Metrics Explained

- **1.8 km Corridor Length:** The physical length of the reference demonstration arterial along Nagar Road, Pune (stretching from Viman Nagar Chowk to Somnath Nagar Chowk).
- **10 Road Segments & 2 Signalized Intersections:** The physical road network topology modeled in the authoritative relational database (`SEG-NR-EB-01` .. `03` , `SEG-NR-WB-01` .. `03` , approach legs, and intersections `INT-VN-01` , `INT-SN-01`).
- **3 Commercial Complexes:** Major electrical consumers modeled along the corridor (Phoenix Marketcity, Solitaire Business Hub, Hyatt Regency) representing over 4.4 MW of commercial load.
- **24/7 Real-Time Telemetry (1.0s to 2.4s):** The ingestion and streaming heartbeat frequency at which physical sensor packets are emitted, validated, and pushed to the console.
- **Before-and-After Slider Metrics:** Displays an average speed increase from 28.5 km/h to 34.2 km/h (+20.0%) and average approach queue reduction from 48 meters down to 28 meters (-41.6%).

2.5 Behind the Scenes: How It Works

The landing page does not query heavy time-series tables directly. It pulls cached metadata from the backend `/api/v1/corridor/summary` endpoint. The before-and-after comparison slider visualizes pre-computed outputs from calibrated SUMO microscopic simulation runs, proving the value of signal coordination before the user logs into the operational console.

2.6 How to Demo This Screen

1. Start at the top: "*Judges, this is our public gateway. It highlights our four core pillars: Traffic Mobility, Microgrid Power, Predictive AI, and Simulation.*"
2. Drag the slider: "*Notice this before-and-after slider. On the left is standard uncoordinated traffic with long queues. On the right, our green-wave coordination allows traffic platoons to progress smoothly.*"
3. Click **"Enter Operations Console"** to transition to Screen 1.

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3. Screen 0B: Digital Twin Provisioning Intake (`#provisioning`)

3.1 Operational Purpose

Proves that the platform is fully generalized and scalable to any city in the world. It provides a formal municipal intake portal where external agencies (e.g. Mumbai Traffic Police, London Transport, Singapore LTA) can upload their road network GeoJSON and register physical sensor streams without touching platform code.

3.2 Technical Stack Used on This Screen

- **React 18 Form State Engine:** Manages multi-field validated intake forms.
- **Pydantic v2 Backend Schemas:** Validates payload geometry, email formats, and sensor bounds at the API boundary (`backend/schemas/provisioning.py`).
- **GeoJSON Parser:** Validates spatial boundary coordinates and polygon topologies.

3.3 UI Controls, Buttons, and Navigation

- **"Submit Provisioning Request":** Validates form fields, shows an animated laser progress bar while transmitting, and POSTs the registration dossier to `/api/v1/provisioning/request`.
- **"Admin Console":** Opens the administrative maker/checker dialog for platform engineers to review and provision pending corridors.
- **"Return to Public Portal" / "Cancel":** Resets form state and returns to `#landing`.
- **"Copy Request ID":** Copies the generated municipal tracking UUID to the clipboard with visual checkmark feedback.
- **"Submit Another Request":** Clears the completed confirmation screen to allow another registration.

3.4 Screen Numbers and Form Fields Explained

- **Target Corridor Bounding Box:** Displays geographical latitude and longitude coordinates defining the spatial envelope of the corridor (e.g. Pune Nagar Road envelope: `[18.552, 73.915]` to `[18.568, 73.935]`).
- **Sensor Count Fields:** Number of planned inductive loop stations (default 10), environmental air stations (default 2), and commercial building sub-meters (default 3).
- **Streaming Interval (Seconds):** Target telemetry broadcast frequency (allowed range: 1.0s to 60.0s; default 2.0s).
- **Tracking ID:** Cryptographic UUIDv4 string (e.g. `REQ-7F9B2C-2026`) assigned to the submission for municipal tracking.

3.5 Behind the Scenes: How It Works

When submitted, the backend creates a provisional corridor registration record in `corridor_registry`. When approved in the admin console, the system dynamically populates the `spatial_entities`, `road_segments`, and `building_zones` tables, instantly spinning up an isolated digital twin instance for that city.

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4. Screen 0C: Municipal Authentication Gateway (`#auth` / `#signin`)

4.1 Operational Purpose

Secures access to municipal command functions using Role-Based Access Control (RBAC). In urban infrastructure governance, junior staff or public observers must never be allowed to authorize traffic signal retiming or commercial power shedding.

4.2 Technical Stack Used on This Screen

- **Cryptographic JWT (JSON Web Tokens):** Employs Python-Jose and BCrypt for stateless authentication.
- **FastAPI OAuth2 Password Flow:** Validates credentials against `users` and `roles` tables.
- **LocalStorage Session Engine:** Stores encrypted session bearer tokens.

4.3 UI Controls, Buttons, and Navigation

- **1-Click Demo Persona Pills:** Four pre-configured quick sign-in buttons for evaluations:
  1. *Traffic Police Officer (Inspector Patil):* Role `TRAFFIC_OPERATOR`. Grants signal optimization and traffic inspection permissions.

2. *Power Grid Operator (Engineer Deshmukh):* Role `GRID_OPERATOR`. Grants microgrid feeder monitoring and load-shedding review permissions.
3. *Environmental Analyst (Dr. Rao):* Role `ENVIRONMENT_ANALYST`. Grants CAAQMS telemetry inspection and anomaly review permissions.
4. *City Commissioner / Mayor (Commissioner Mehta):* Role `COMMISSIONER`. Super-user role with complete authority to formally approve advisory dossiers.

- **"Sign In" vs "Create Account" Tabs:** Toggles between standard credential login and new agency registration.
- **"Show / Hide Password":** Toggles input field visibility for security.
- **"Back to Overview":** Returns to `#landing`.

4.4 Behind the Scenes: How It Works

Upon clicking any persona or submitting credentials, the backend generates a signed JWT token containing the user identity, assigned municipal agency, and RBAC permission scopes. The token has an expiration TTL of 60 minutes. The frontend stores this token and includes it in the `Authorization: Bearer <token>` header for all subsequent REST mutations and WebSocket connections.

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5. Screen 1: Corridor Operations Console (`#operations`)

5.1 Operational Purpose

The primary tactical command canvas of the platform. It provides operators with a synchronized 2D/3D operational picture of the entire corridor, showing live speeds, signal states, power loading, and incoming machine learning forecasts.

5.2 Technical Stack Used on This Screen

- **2D Tactical GIS:** Leaflet 1.9 rendering GeoJSON road segment polylines with dynamic speed-based color coding (green, yellow, red).
- **3D Digital Twin Viewer:** CesiumJS 1.120 rendering photogrammetric terrain, 3D extruded commercial buildings, and animated vehicles moving at speeds matching physical sensor readings.
- **Real-Time WebSocket Fabric:** `/ws/operations` pushing JSON-LD delta updates every 2.0 seconds.
- **Recharts 2.12:** Powers the multi-horizon forecasting charts inside the slide-out drawer.

5.3 UI Controls and Buttons Explained

- **View Mode Switcher Toolbar (Top-Right of Map):**
  - *2D Map Button:* Toggles the lightweight tactical Leaflet GIS map.
  - *3D Twin Button:* Toggles the immersive CesiumJS 3D photogrammetric twin viewer.
  - *Table Button:* Opens a WCAG 2.1 AA accessible data table showing all road segments, current speeds, flows, and LOS grades in tabular form for screen readers.
- **Basemap Style Selector (Under Map Toolbar):**
  - *Dark Button:* Switches map background to dark navy command canvas (optimal for night shifts).
  - *Satellite Button:* Displays high-resolution satellite imagery tiles showing physical road markings and building rooftops.
  - *Streets Button:* Displays clean vector street map lines with neighborhood names.
- **3D Camera Navigation Controls:** Floating overlay with Pitch, Heading, Fly-to-Overview, and Reset Camera triggers.
- **Segment / Entity Click Trigger:** Clicking any road segment polyline or building polygon opens the **Entity Detail Drawer** on the right side of the screen.

5.4 Screen Numbers and Metrics Explained

- **Top Corridor Metrics HUD (Floating Top-Left Card):**
  - **Corridor Average Speed (e.g. `32.4 km/h`):** The harmonic space-mean speed across all 10 monitored segments.
  - **Speed Trend Delta (e.g. `-3.2 km/h`):** Velocity change compared to the rolling 60-minute historical average.
  - **Total Vehicular Flow (e.g. `1,840 veh/h`):** Aggregate throughput rate of vehicles crossing active detector stations.
  - **Commercial Power Demand (e.g. `4.42 MW`):** Real-time aggregate electrical load drawn by corridor commercial facilities.
  - **Corridor AQI (e.g. `135 AQI Moderate`):** Live air quality index from on-corridor sensors.
  - **Provenance Badge (e.g. `LIVE` green / `REPLAY` blue):** Mandatory metadata tag stating data origin.
- **Entity Detail Drawer (Slide-Out Right Panel):**
  - **Average Speed:** Segment velocity in km/h.
  - **Vehicle Flow:** Segment hourly throughput rate in veh/h.
  - **Occupancy:** Pavement detector occupancy percentage (`%`).
  - **Queue Length:** Stopline vehicular queue length in meters (`m`).
  - **Congestion Index (0.000 to 1.000):** Dimensionless speed deficit index: `CI = max(0, 1 - (speed / speed_limit))`.
  - **Level of Service (LOS A through F):** Qualitative road performance grade.

- **Multi-Horizon Speed Forecast:** Displays predicted speeds at 15m, 30m, and 60m horizons with shaded Conformal 80% and 90% uncertainty bands.
- **TreeSHAP Top-5 Feature Contributions:** Waterfall chart showing exactly which features caused the prediction (e.g. 15-minute speed trend `-4.2 km/h`, evening rush multiplier `-3.8 km/h`).

5.5 Behind the Scenes: How It Works

When the screen mounts, it establishes a persistent WebSocket connection to `/ws/operations`. Every 2 seconds, the backend `TelemetryStreamerWorker` broadcasts an enriched JSON-LD packet. If network latency delays a packet, the client-side telemetry heartbeat maintains realistic micro-fluctuations so operator screens never appear frozen.

5.6 How to Demo This Screen

1. Point to the top HUD: "*Here is the live corridor heartbeat: 32.4 km/h average speed, 1,840 vehicles per hour, and 4.42 MW of commercial power.*"
2. Toggle the 3D Twin button: "*With one click, we switch from 2D tactical GIS into our 3D Cesium Digital Twin, rendering photogrammetric terrain and animated vehicles.*"
3. Click segment `SEG-NR-EB-01` to open the drawer: "*Notice our AI forecast: speed will drop to 28 km/h in 30 minutes. TreeSHAP shows the top-5 reasons why, and Conformal prediction provides guaranteed 90% confidence bands.*"

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6. Screen 2: Traffic Analytics & Near-Term Forecasting (`#traffic`)

6.1 Operational Purpose

Provides traffic engineers and police officers with detailed capacity analysis, signal phase timing countdowns, and automated Webster signal timing optimization for congested intersections.

6.2 Technical Stack Used on This Screen

- **Indo-HCM (IRC:106) Dual-Criteria Engine:** Evaluates road capacity under heterogeneous Indian mixed-traffic conditions.
- **Webster Signal Timing Solver:** Solves F.V. Webster minimum-delay equations in Python/TypeScript.
- **SPaT Adapter:** Virtual Signal Phase and Timing controller adapter broadcasting active phase states.

6.3 UI Controls and Buttons Explained

- **Policy Standard Toggle Buttons (Top-Right):**
  - *Indo-HCM (IRC:106) Button:* Activates Indian Highway Capacity Manual rules, converting mixed traffic into Passenger Car Units (PCU) and applying dual-criteria evaluation on speed and Volume-to-Capacity (`V/C`) ratio.
  - *US HCM 6th Ed Button:* Switches evaluation to standard US Highway Capacity Manual tables assuming homogeneous passenger car lanes.
- **"Compute Webster Signal Optimization" / "Webster Plan" Button:** Opens the interactive Webster Timing Plan modal to recalculate optimal cycle lengths and green splits for Viman Nagar Chowk (`INT-VN-01`).
- **"Inspect" Segment Buttons:** Focuses the analytics charts on a specific road segment.

6.4 Screen Numbers and Metrics Explained

- **Volume-to-Capacity (`V/C`) Ratio Gauge (e.g. `0.78`):** Measures road saturation degree. Values below 0.65 represent free/stable flow; values between 0.80 and 1.00 represent near-capacity conditions; values above 1.00 indicate oversaturated breakdown (LOS F).
- **Signal Controller Phase Card:**
  - **Active Phase (e.g. `Phase 1: East-West Arterial`):** The signal movement currently receiving green right-of-way.
  - **Countdown Timer (e.g. `18s`):** Seconds remaining before the phase transitions to yellow and all-red clearance.
  - **Cycle Length (e.g. `120s`):** Total duration of one complete rotation through all signal phases.
  - **Controller Mode (e.g. `COORDINATED` / `TIME_OF_DAY`):** Current control mode.
- **Webster Optimization Modal Numbers:**
  - **Total Lost Time (`L = 14s`):** Sum of clearance intervals and startup delays across phases.
  - **Critical Volume Ratio (`Y = 0.75`):** Sum of approach volume-to-saturation flow ratios (`Y = sum(q_i / s_i)`).
  - **Optimum Cycle Length (`C_0 = 104s`):** Computed via `C_0 = (1.5 * L + 5) / (1 - Y) = (1.5 * 14 + 5) / (1 - 0.75) = 26 / 0.25 = 104s`.
  - **Recommended Green Splits:** Phase 1 (Nagar Road) = 48s; Phase 2 (Viman Nagar cross street) = 42s.

6.5 Behind the Scenes: How It Works

The Indo-HCM engine weights 2-wheelers at 0.5 PCU, auto-rickshaws at 1.0 PCU, cars at 1.0 PCU, and buses at 3.0 PCU. The Webster solver recalculates cycle lengths in real time based on active approach sensor counts, mathematically minimizing total vehicle delay at the stopline.

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7. Screen 3: Energy Analytics & Microgrid Integration (`#energy`)

7.1 Operational Purpose

Couples municipal road traffic with the electrical distribution grid. It monitors 3-phase AC power physics, rooftop solar generation, battery storage (BESS), and 22kV feeder thermal headroom to prevent transformer overload during peak hours.

7.2 Technical Stack Used on This Screen

- **3-Phase AC Physics Engine:** Computes real active power, reactive power, apparent power, and power factor.
- **Kirchhoff Conservation Engine:** Verifies microgrid energy balance across grid import, solar, batteries, and loads.
- **Recharts Diurnal Load Profiler:** Plots 24-hour commercial electrical demand schedules.

7.3 UI Controls and Buttons Explained

- **Sub-Tab Navigation Buttons:**
  - *Electrical Network Topology & 3-Phase Telemetry Button:* Displays the single-line electrical distribution tree and live substation meter readings.
  - *Commercial Demand Forecasting & Diurnal Schedules Button:* Displays 24-hour commercial load profiles and solar generation curves.
- **"Demand-Response Dossier" Button:** Evaluates building load shedding opportunities and generates candidate peak-shaving recommendations.
- **"Advance Step (+60s)" Button:** Steps the background energy simulation forward by 60 seconds to demonstrate load changes.
- **"Auto-Advance Off / Auto-Advancing (10s)" Preset Pill:** Toggles continuous background simulation advancement every 10 seconds.

7.4 Screen Numbers and Metrics Explained

- **3-Phase AC Telemetry Card:**
  - **Line-to-Line Voltage (`V_LL = 415 V`):** Secondary distribution voltage delivered to commercial main boards.
  - **Phase Current (`I_L = 3600 A`):** RMS line current drawn per phase.
  - **Active Real Power (`P = 2484 kW`):** Usable power doing work: `P = sqrt(3) * V_LL * I_L * cos(phi)` .
  - **Reactive Power (`Q = 724 kVAR`):** Inductive electromagnetic power: `Q = sqrt(3) * V_LL * I_L * sin(phi)` .
  - **Apparent Power (`S = 2587 kVA`):** Vector sum of delivered power: `S = sqrt(P^2 + Q^2)` .
  - **Power Factor (`PF = 0.96`):** Electrical efficiency ratio (`PF = P / S`). Utilities penalize values below 0.95.
- **Feeder Thermal Loading Card:**
  - **Ampacity Utilization (`76.5%`):** Conductor current loading against the 800A rated thermal limit of the 22kV cable.
  - **Available Electrical Headroom (`694 kW`):** Remaining commercial power that can be drawn before hitting thermal warning capacity: `Headroom = sqrt(3) * V * (I_rated - I) * PF` .
- **Distributed Energy Resources (DER) Card:**
  - **Rooftop Solar PV (`1200 kW`):** Active generation from commercial solar panels.
  - **BESS Battery Power (`-300 kW`):** Battery power flow (negative indicates charging from surplus solar; positive indicates discharging to shave grid peaks).
  - **State of Charge (`78.5% SoC`):** Remaining stored energy capacity in the battery system.

7.5 Behind the Scenes: How It Works

The microgrid engine enforces Kirchhoff energy conservation: `P_grid + P_solar + P_bess = P_loads + P_losses` . When commercial shopping complexes approach maximum demand contract limits (e.g. 6.8 MW at Phoenix Marketcity), the engine warns operators before substation transformers overheat.

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8. Screen 4: Environmental Context & Air Quality (`#environment`)

8.1 Operational Purpose

Monitors the human and public health impact of traffic congestion. It tracks ambient criteria pollutants from Continuous Ambient Air Quality Monitoring Stations (CAAQMS), computes official CPCB AQI scores, and detects abnormal pollution surges using machine learning.

8.2 Technical Stack Used on This Screen

- **CPCB Linear Interpolation Solver:** Implements official Central Pollution Control Board (India) AQI formulas.
- **Scikit-Learn Isolation Forest:** Unsupervised machine learning model detecting air quality anomaly spikes.
- **Open-Meteo Weather API Integration:** Ingests live ambient temperature, humidity, and wind vectors.

8.3 UI Controls and Buttons Explained

- **Station Selector Dropdown:** Allows the operator to switch between monitoring stations:

1. *Dual Arterial Corridor Micro-Climate Station (On-Corridor):* Immediate roadside sensor tracking vehicle exhaust.
2. *Airport Weather & CAAQMS Station:* Regional background meteorological reference station.
3. *Central Sector Regional CAAQMS (MPCB Station #MH004):* Regional regulatory station.

- **"Retry" Error Button:** Triggers immediate reconnection if weather API fails.

8.4 Screen Numbers and Metrics Explained

- **Air Quality Index Gauge (e.g. `135 AQI Moderate`):** The composite index calculated via official CPCB piecewise linear interpolation: `I = I_low + ((I_high - I_low) / (C_high - C_low)) * (C - C_low)`.
- **Pollutant Concentration Cards:**
  - **PM2.5 (`48.5 ug/m3`):** Fine inhalable particles <= 2.5 microns from combustion engines.
  - **PM10 (`92.0 ug/m3`):** Coarse road dust and tire/brake wear particles.
  - **NO2 (`36.0 ug/m3`):** Nitrogen dioxide gas emitted by diesel vehicles.
  - **CO (`410 ug/m3` / `0.41 mg/m3`):** Carbon monoxide gas from incomplete fuel combustion.
- **Meteorological Vector Card:**
  - **Temperature (`28.5 deg C`) & Relative Humidity (`56%`):** Controls air buoyancy and thermal comfort.
  - **Wind Speed (`11.2 km/h`) & Direction (`WSW 240 deg`):** Atmospheric vector controlling exhaust plume dispersion.
- **Statistical Anomaly Detection Card:**
  - **Isolation Forest Score (`0.38`):** Measures tree isolation depth `s(x, n) = 2^(-E(h(x)) / c(n))`. Values below 0.60 are normal; values above 0.60 trigger instant anomaly warnings.
  - **Diurnal Z-Score (`+0.29`):** Number of standard deviations from historical time-of-day mean (`Z = (x - mu) / sigma`).

8.5 Behind the Scenes: How It Works

Vehicular emissions accumulate when arterial traffic crawls below 18 km/h. The Isolation Forest runs on every incoming packet to distinguish normal commuter peak emissions from dangerous industrial discharge spikes or sensor chamber dust contamination.

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9. Screen 5: Scenario Studio Simulation Sandbox (`#scenarios`)

9.1 Operational Purpose

The what-if experimentation laboratory of the digital twin. City engineers test signal retiming, green waves, and commercial load shedding in software using microscopic physics before implementing changes on physical streets.

9.2 Technical Stack Used on This Screen

- **Eclipse SUMO 1.20 (Simulation of Urban MObility):** Microscopic car-following and lane-changing physics engine.
- **Python TraCI Library:** Asynchronous command interface communicating with the SUMO simulation daemon.
- **Kinematic Fallback Runner (`simulation/runner.py`):** In-process kinematic simulation runner guaranteeing 100% execution even if external C++ SUMO binaries are unavailable.

9.3 UI Controls, Buttons, and Sliders Explained

- **Approved Intervention Template Selector:** Dropdown offering verified municipal scenario templates:
  - `SCEN-INT-01`: *Arterial Green Wave Coordination (Viman Nagar Chowk to Somnath Nagar Chowk).*
  - `SCEN-INT-02`: *Commercial Peak Shaving and BESS Microgrid Dispatch.*
  - `SCEN-INT-03`: *Emergency Vehicle Corridor Preemption & Incident Lane Diversion.*
- **Slider 1: Dynamic Parameter Slider (Context-Aware):**
  - For Green Wave (`SCEN-INT-01`): Adjusts arterial green extension from `+0s` to `+30s` with quick-preset buttons (`+5s Conservative`, `+15s Calibrated`, `+25s Aggressive`).
  - For Coordination Offset (`SCEN-INT-02`): Adjusts progression offset from `0s` to `60s` with presets (`25s Tight`, `35s Optimal Wave`, `50s Wide`).
- **Slider 2: Traffic Demand Multiplier:** Scales vehicular arrival flow rate from `0.60x` (night off-peak) to `1.40x` (severe festival surge). Presets: `0.80x Off-Peak`, `1.00x Evening Peak`, `1.30x Severe Surge`.
- **Slider 3: Random Simulation Seed:** Sets the pseudorandom seed for stochastic driver behavior variance. Includes a **"Random"** button to generate random seeds.
- **"Execute Comparative SUMO Simulation" Button:** Launches the comparative simulation run. Displays a loading spinner: "*Simulating Corridor Micro-flows...*" while SUMO processes vehicle kinematic steps.
- **Recent Simulation Runs History List (Left Sidebar):** Lists previous execution runs with run IDs, timestamps, and delay deltas. Clicking any run instantly loads its comparative results.
- **"Propose Advisory Intervention from this Simulation" Button:** Appears after a simulation completes with positive results. Opens the advisory modal to submit the tested parameters as a candidate recommendation.

9.4 Screen Numbers and Comparative Results Explained

- **Comparative KPI Results Table (Baseline vs Intervention):**
  - **Average Corridor Velocity:** Baseline `28.4 km/h` vs Simulated Intervention `32.6 km/h` (Net Velocity Gain: `+4.2 km/h` or `+14.8%`).
  - **Average Stopline Queue Length:** Baseline `52.0 meters` vs Intervention `37.4 meters` (Queue Reduction: `-28.1%`).
  - **Total Commuter Delay Saved:** Cumulative vehicle-hours saved across all corridor commuters (e.g. `185 vehicle-hours`).
  - **Estimated Fuel Consumption Saved:** Volume of fuel saved by avoiding stop-and-go acceleration cycles (e.g. `142 liters / hour`).
  - **Emissions Reduction:** Metric tons of CO2 abated based on fuel savings (e.g. `-0.33 tCO2 / day`).

9.5 Behind the Scenes: How It Works

SUMO runs individual vehicle agent kinematics using the Krauss car-following model. It runs two passes: first with baseline timings, then with the operator adjusted parameters. All results are tagged strictly as `SIMULATION` and stored in `simulation_runs`, preserving the State Separation Invariant (simulations never overwrite ground-truth data).

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10. Screen 6: Governance & Advisory Center (`#recommendations`)

10.1 Operational Purpose

The human-in-the-loop governance hub. It enforces the **Strict Non-Actuation Invariant**, ensuring that AI models never autonomously control physical streetlights or substation breakers. Instead, certified city officials review, evaluate, and formally authorize structured recommendations.

10.2 Technical Stack Used on This Screen

- **Rule-Based Advisory Engine (`backend/services/advisory_engine.py`):** Synthesizes multi-domain observations into candidate intervention dossiers.
- **Cryptographic Audit Logger:** Records all approvals with officer identities, timestamps, and SHA-256 verification hashes.
- **Role-Based Authorization Guards:** Enforces permission scopes (only `OPERATOR` and `COMMISSIONER` roles can authorize plans).

10.3 UI Controls and Buttons Explained

- **Tab Mode Navigation Buttons:**
  - *Active Advisory Dossiers Button:* Shows pending, approved, and active recommendations.
  - *Municipal Dispatch Orders & Audit Trail Button:* Displays the immutable permanent log of all historical municipal interventions.
- **"Re-Evaluate Rules" Button:** Manually forces the advisory engine to re-scan current corridor telemetry and generate fresh recommendations immediately.
- **Domain Filter Pills:** Filters dossiers by domain: `ALL`, `TRAFFIC`, `ENERGY`, `ENVIRONMENT`.
- **Status Filter Pills:** Filters dossiers by lifecycle status: `ALL`, `PENDING_APPROVAL`, `APPROVED`, `REJECTED`, `EXECUTED`, `SUPERSEDED`.
- **"Reset All Filters" Button:** Clears all active filters to view all records.
- **Dossier Action Buttons (On Each Advisory Card):**
  - *Authorize Advisory Plan (Green Button):* Formally approves the dossier, updating status to `APPROVED` and writing an entry to the audit log.
  - *Reject Plan (Red Button):* Rejects the recommendation with a required justification note.
- **"Refresh Log" Button (In Audit Tab):** Reloads the latest SHA-256 audit log records from the database.

10.4 Screen Numbers and Advisory Card Elements Explained

- **Advisory Summary KPI Counters (Top Banner):**
  - **Active Dossiers Count:** Number of actionable proposals currently generated.
  - **Pending Review Count:** Proposals awaiting officer inspection.
  - **Approved Today:** Count of authorized plans executed during the current shift.
- **Detailed Advisory Dossier Card Structure:**
  - **Dossier Identifier:** e.g. `ADV-2026-TRF-0842`.
  - **Target Infrastructure Asset:** e.g. `urn:ngsi-ld:Intersection:PUNE:INT-VN-01`.
  - **Action Type:** e.g. `EXTEND_PHASE_GREEN` or `COMMERCIAL_LOAD_SHED`.
  - **Recommended Parameters:** e.g. `Extend East-West Green by +15s (Phase 1 Split: 50s)`.
  - **Predicted Impact:** Speed increase `+3.8 km/h`, queue reduction `-24%`, power headroom `+120 kW`.
  - **Model Confidence Score:** e.g. `92.4%` (derived from Conformal Prediction uncertainty bands).
  - **Audit Record Row:** Displays the unique Dispatch Order ID, Approving Officer Name, Assigned Role, Timestamp, and SHA-256 Checksum Hash.

10.5 Behind the Scenes: How It Works

The advisory engine runs rule templates every 60 seconds. For example, if arterial speed drops below 20 km/h and upstream queue exceeds 50 meters, it triggers Rule `RULE-TRF-GW-01`. Crucially, the platform stops at proposal: field actuation requires a human click, ensuring zero public liability from AI hallucinations.

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11. Screen 7: Pilot Evaluation & Executive Reporting (`#evaluation`)

11.1 Operational Purpose

Provides municipal commissioners, auditors, and evaluators with scientific proof that the AI models are accurate and reliable. It benchmarks machine learning forecasts against chronological holdout data, explains forecast skill scores, and generates downloadable executive briefings.

11.2 Technical Stack Used on This Screen

- **Chronological Holdout Evaluation Engine (`backend/services/evaluation_service.py`):** Evaluates models on the final 10% chronological holdout dataset with zero lookahead data leakage.
- **Markdown Executive Report Compiler:** Compiles dynamic multi-domain telemetry into structured executive briefings.
- **File Export Streamer:** Downloads raw markdown reports directly to the user machine.

11.3 UI Controls and Buttons Explained

- **Tab Selector Buttons (Top Navigation):**
  - *Evaluation Benchmarks Tab:* Displays forecast horizon cards and the chronological holdout comparison table.
  - *Executive Briefing Tab:* Displays the complete, formatted executive evaluation report.
  - *Data Provenance Audit Tab:* Displays the data integrity table tracking observations, calibrations, and sensor origins.
- **Briefing View Mode Toggle Buttons:**
  - *Formatted Report Button:* Renders the executive report with formatted headers, tables, and metric badges.
  - *Raw Markdown Button:* Displays raw Markdown code with syntax highlighting for technical inspection.
- **"Export Executive Briefing (.md)" Button:** Triggers an immediate browser download of the complete executive briefing document (`digital_twin_executive_briefing.md`).
- **"Copy Markdown" Button:** Copies the entire markdown report to the system clipboard, displaying a temporary green checkmark: "*COPIED*".

11.4 Screen Numbers, Tables, and Key Terms Explained

The Three Forecast Horizon Cards:

1. **15m Horizon Speed Card:**
   - **Model MAE:** `2.45 km/h` (Mean Absolute Error of our XGBoost model).
   - **Baseline MAE:** `3.25 km/h` (Mean Absolute Error of the standard persistence baseline).
   - **Forecast Skill Score:** `0.245` (Standardized skill score).
   - **Gain vs Base Badge:** `+24.5%` (Green improvement badge).
2. **30m Horizon Speed Card:**
   - **Model MAE:** `3.12 km/h`. Baseline MAE: `3.89 km/h`.
   - **Forecast Skill Score:** `0.198`.
   - **Gain vs Base Badge:** `+19.8%`.
3. **60m Horizon Speed Card:**
   - **Model MAE:** `3.95 km/h`. Baseline MAE: `4.60 km/h`.
   - **Forecast Skill Score:** `0.141`.
   - **Gain vs Base Badge:** `+14.1%`.

Conformal Coverage Card:

- **Empirical Coverage (`90.2%`):** The actual percentage of holdout observations that fell within the model's 90% Conformal confidence interval, verifying that the uncertainty bounds are mathematically valid.

The Holdout Benchmark Table:

- **Split Ratio (`80/10/10 Chronological Split`):** Proves that the data was split strictly in time order (first 80% training, next 10% validation, final 10% holdout testing), preventing lookahead data leakage.
- **Column 1: Forecast Horizon:** 15-Minute, 30-Minute, and 60-Minute horizons.
- **Column 2: Persistence Baseline:** The prediction error if one simply assumes that future traffic speed will be identical to the current speed.
- **Column 3: XGBoost Model:** The actual prediction error achieved by our machine learning model.
- **Column 4: Skill Score:** Standardized forecast quality metric.

- **Column 5: Gain vs Base:** The exact percentage reduction in error achieved over the baseline.

11.5 Deep Technical Concepts Explained: What the Terms Mean

What "Persistence Baseline" Means:

In time-series forecasting, a **Persistence Baseline** (or naive baseline) is the simplest possible prediction: it assumes that the future value will be exactly equal to the most recently observed value:

```
y_predicted(t + h) = y_observed(t)
```

For example, if speed right now is 32 km/h, the persistence baseline predicts that speed 30 minutes from now will also be 32 km/h. In professional transport engineering, any machine learning model that cannot consistently beat the persistence baseline is considered useless because it fails to add intelligence over a zero-cost guess.

What "Gain vs Base" Means:

Gain vs Base (Gain versus Baseline) measures the exact percentage reduction in prediction error achieved by our XGBoost machine learning model compared to the persistence baseline:

```
Gain_vs_Base = ((Persistence_MAE - Model_MAE) / Persistence_MAE) * 100%
```

Worked Numerical Example (15-Minute Horizon):

- Persistence Baseline MAE = `3.25 km/h`.
- XGBoost Model MAE = `2.45 km/h`.
- `Gain_vs_Base = ((3.25 - 2.45) / 3.25) * 100% = (0.80 / 3.25) * 100% = +24.6%` (displayed as `+24.5%` due to rounding).
- Meaning: Our XGBoost model eliminates nearly one-quarter of the error that a standard baseline makes, proving significant predictive power.

What "Forecast Skill Score" Means:

The **Forecast Skill Score** is a standardized metric borrowed from atmospheric and transport science that quantifies forecast accuracy relative to a reference benchmark:

```
Skill_Score = 1 - (Model_MAE / Persistence_MAE)
```

- If `Skill_Score = 0.0`: The model is no better than guessing the current number (zero skill).
- If `Skill_Score < 0.0`: The model is worse than the baseline (negative skill, model discarded).
- If `Skill_Score > 0.0`: The model demonstrates genuine forecasting skill.
- Our 15-minute horizon achieves a Skill Score of `0.245`, proving strong statistical outperformance.

Why XGBoost is Used Here:

Rather than using an opaque deep neural network, the platform employs **XGBoost (Extreme Gradient Boosted Trees)** because:

1. It handles tabular time-series lag features, cyclical sine/cosine time features, and discrete weather categories with high accuracy.
2. It operates efficiently in sub-millisecond inference time on municipal servers without expensive GPUs.
3. It integrates natively with **TreeSHAP**, allowing every forecast to be mathematically decomposed into exact feature attributions.

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12. Screen 8: System & Data Health Observability (`#health`)

12.1 Operational Purpose

The IT and infrastructure health cockpit. It monitors system telemetry streams, database query latencies, active relational tables, and inspects the Data Quarantine Queue to ensure zero bad data enters the platform.

12.2 Technical Stack Used on This Screen

- **FastAPI Health & Telemetry Observer:** Collects in-process server statistics, memory usage, and background worker status.
- **SQLAlchemy Schema Inspector:** Audits the 29 relational tables and foreign key indexes.
- **Pydantic Bounds Firewall:** Validates incoming telemetry packets and populates the quarantine queue.

12.3 UI Controls and Buttons Explained

- **"Refresh" Button:** Reloads health metrics and database query round-trip times immediately.
- **"Inspect Queue" Button:** Opens the **Data Quarantine Queue Modal**, allowing engineers to inspect rejected sensor packets.
- **Telemetry Streamer Action Buttons:**
  - *"Tick Now" Button:* Manually forces the background streamer worker to execute one immediate telemetry cycle.
  - *"Pause / Resume" Button:* Pauses the autonomous streaming worker to freeze telemetry for debugging, or resumes streaming.
- **"Inspect Manifest" Button:** Opens the corridor asset manifest modal, listing every registered spatial URN and its configuration.
- **Quarantine Reason Filter Pills (In Modal):** Filters rejected packets by failure reason:
  - `ALL`: Shows all quarantined records.
  - `SPEED_EXCEEDS_LIMIT`: Packets reporting impossible speeds (> 120 km/h).

- `NEGATIVE_POWER`: Electrical meters reporting negative power consumption.
- `FUTURE_TIMESTAMP`: Sensor packets stamped with timestamps ahead of server time.
- `SCHEMA_MALFORMED`: Corrupted or invalid JSON payloads.

- **"Close Queue Inspector" Button:** Closes the modal.

12.4 Screen Numbers and Health Cards Explained

- **Overall System Status (e.g. `HEALTHY` Green Badge):** Confirms all services, background streaming tasks, and database connections are operational.
- **Database Round-Trip Latency (e.g. `2.4 ms`):** The average time taken to execute a standard read/write query against the active database.
- **Active Table Count (`29 Tables Created & Active`):** The complete count of authoritative database tables actively persisting twin data.
- **Telemetry Streamer Telemetry:**
  - *Ticks Count (e.g. `1,420 ticks`):* Number of simulation/streaming cycles executed since server boot.
  - *Interval (`2.0s`):* Streaming update cadence.
  - *Source Mode (`REPLAY` / `SIMULATION`):* Active streamer mode.
- **Ingestion Firewall Statistics:**
  - *Total Ingested Packets (e.g. `42,850`):* Total telemetry packets processed.
  - *Total Quarantined (e.g. `14 rejected`):* Erroneous packets caught and isolated (`0.03%` error rate).
- **Server Resources:** Python ASGI process RSS memory (e.g. `142 MB`) and active WebSocket client count.

12.5 Behind the Scenes: How It Works

Before any telemetry packet touches an observation table, it passes through the Pydantic Ingestion Firewall. If a faulty sensor transmits an impossible speed of 380 km/h or a negative power reading of -500 kW, the packet is rejected, tagged `INVALID`, and written to `data_quarantine`. This protects machine learning models from training on garbage data.

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13. Summary Matrix of All 11 Screens

Screen #	Route Hash	Screen Name	Primary User Persona	Core Technological Highlight
0	`#landing`	Public Landing Page	Public & Commissioners	Interactive before/after diorama and corridor slider
0B	`#provisioning`	Provisioning Intake	Municipal Onboarding Staff	Dynamic GeoJSON parsing and spatial entity provisioning
0C	`#auth`	Municipal Auth Gateway	All Municipal Officers	JWT authentication with 4 one-click demo personas
1	`#operations`	Corridor Operations Console	Field Traffic Operators	Dual 2D Leaflet & 3D Cesium photogrammetric twin viewer
2	`#traffic`	Traffic Analytics & Forecasting	Traffic Police & Engineers	Indo-HCM (IRC:106) evaluation & Webster signal timing
3	`#energy`	Energy Analytics & Microgrid	Grid Operators & Utilities	3-phase AC Kirchhoff balance & feeder thermal headroom
4	`#environment` | Environmental Context	Pollution Control Boards	Official CPCB AQI & Isolation Forest anomaly detection	
5	`#scenarios`	Scenario Studio Sandbox	Planners & Simulation Staff	SUMO microscopic car-following & green wave what-if
6	`#recommendations`	Governance & Advisory Center	Certified Approving Officers	Strict Non-Actuation & SHA-256 tamper-evident audit log
7	`#evaluation`	Pilot Evaluation & Reporting	Auditors & Commissioners	Chronological holdout benchmarks, Gain vs Base, Skill Score
8	`#health`	System Health Observability	IT Systems Administrators	Pydantic firewall, quarantine queue, & streamer controls